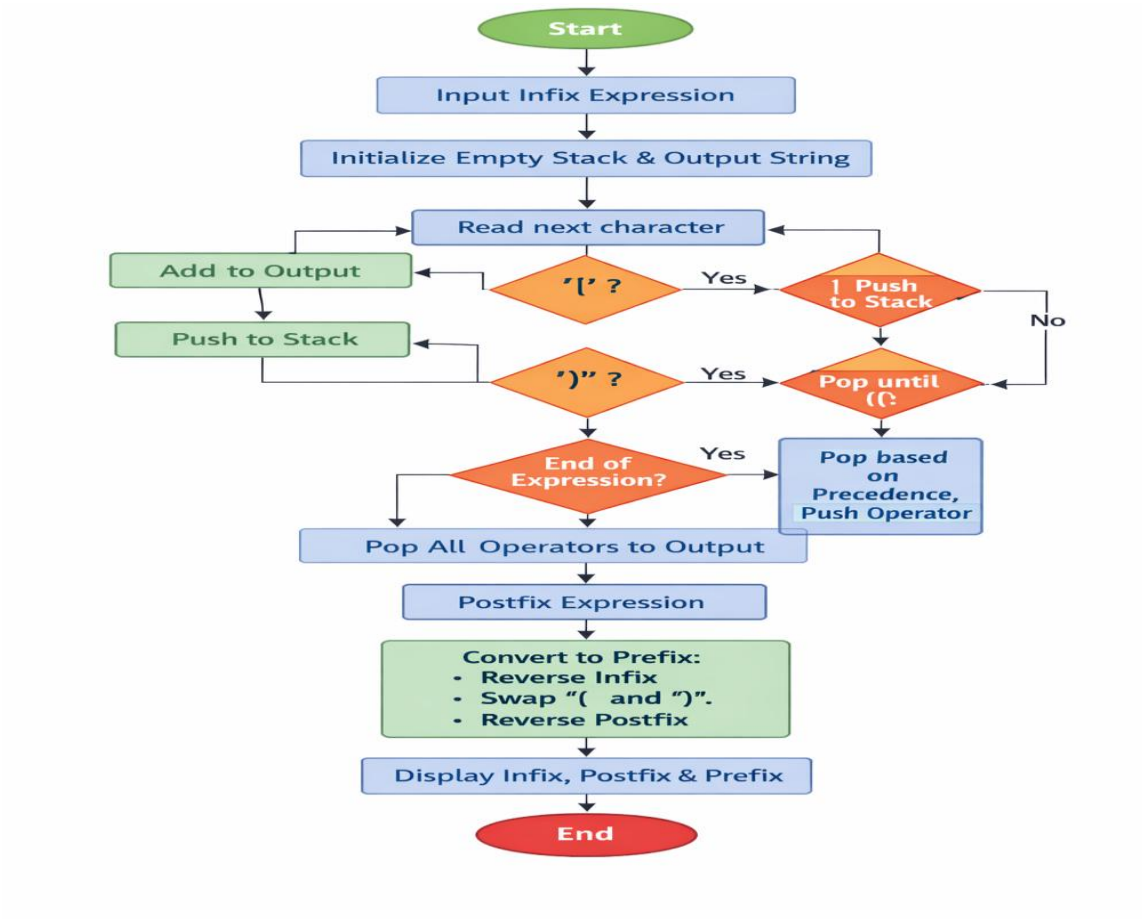
Assignment Title:

Conversion of Infix Expression to Postfix and Prefix Using Java

Abstract

This assignment demonstrates how to convert an infix expression into postfix and prefix forms using Java. It includes a detailed workflow represented by a flowchart, pseudocode outlining the algorithm, a complete Java program implementation, and example outputs showing the conversion results.

1. WorkFlow



2. Pseudocode

function infixToPostfix(expression):

```
    create empty stack
    create empty output string
    for each character in expression:
        if operand → add to output
        if '(' → push to stack
        if ')' → pop until '('
        if operator → pop higher/equal precedence operators, then push current
    pop remaining operators to output
    return output


function infixToPrefix(expression):
    reverse expression
    swap '(' and ')'
    postfix = infixToPostfix(reversed expression)
    prefix = reverse(postfix)
    return prefix
```

3.Example Conversion

Expression: (A-B/C)*(A/K-L)

- (→ push
- A → output → A
- - → push
- B → output → AB
- / → push (higher precedence than -)
- C → output → ABC
-) → pop until ‘(‘→ output becomes ABC/-
- * → push
- (→ push
- A → output → ABC/-A

- / → push
- K → output → ABC/-AK
- - → push
- L → output → ABC/-AKL
-) → pop until (→ output becomes ABC/-AK/L-
- End: pop remaining * → final postfix = ABC/-AK/L-*
- Reverse logic → prefix = *-A/BC-/AKL

5. Example Output (from VS Code terminal)

Infix: (A-B/C)*(A/K-L)

Postfix: ABC/-AK/L-*

Prefix: *-A/BC-/AKL